

PrimeTrade Market Sentiment Analysis Report

Bitcoin Fear & Greed Index vs. Hyperliquid Trader Performance

Author: Sidhartha Kumar Malla | **Repository:** PrimeTrade-Sentiment-Trading-Analysis

Objective

The primary goal of this project is to analyze the relationship between Bitcoin market sentiment and trader performance on the Hyperliquid platform. Market sentiment, driven by collective investor psychology, often dictates trading behavior and can significantly influence profitability, trade direction, and risk appetite. This project quantifies that influence by combining external sentiment data with real historical trading records.

Key questions addressed:

- How trader profitability changes during Fear versus Greed periods
- Whether Long or Short positions perform better under different market conditions
- Which assets generate the highest profits across sentiment states
- How trader behavior (position sizing, frequency) shifts with market psychology
- Actionable insights for building sentiment-aware trading strategies

Dataset Description

Two primary datasets power this analysis:

Dataset	Source	Key Fields
Bitcoin Fear & Greed Index	Alternative.me Crypto API	Date, Sentiment Label (Extreme Fear / Fear / Neutral / Greed / Extreme Greed)
Hyperliquid Trader Data	Hyperliquid Exchange	Account, Coin, Execution Price, Size Tokens, Size USD, Side, Timestamp, Direction, Closed PnL

Bitcoin Fear & Greed Index aggregates market volatility, trading volume, social media sentiment, Bitcoin dominance, and Google Trends data into a single daily classification score.

Hyperliquid Historical Trader Data captures individual trade-level records including realized profit/loss (Closed PnL), allowing granular analysis of trader outcomes aligned to each sentiment date.

Data Cleaning

Before analysis, the raw datasets underwent a structured cleaning pipeline to ensure consistency and reliability:

- **Timestamp Conversion:** All timestamp fields standardized to datetime objects for temporal alignment
- **Date Extraction:** Trading dates extracted from full timestamps to serve as the merge key
- **Missing Value Handling:** Null and incomplete records addressed via imputation or removal
- **Sentiment Label Standardization:** Fear & Greed classifications normalized for uniform categorical values
- **Duplicate Removal:** Duplicate trade records identified and eliminated
- **Data Type Validation:** Numeric fields (Closed PnL, Size USD) verified and cast appropriately

Methodology

Data Integration

The cleaned trading data and Fear & Greed Index were merged using trading dates as the common key. This created a unified dataset where every trade record was tagged with the prevailing market sentiment for that day, enabling direct comparison of outcomes across sentiment states.

Exploratory Data Analysis (EDA)

The integrated dataset was analyzed across seven core dimensions:

Analysis	Description
Profitability by Sentiment	Aggregate Closed PnL grouped by Fear & Greed classification
Win Rate by Sentiment	Percentage of profitable trades per sentiment condition
Trade Size by Sentiment	Average position sizing during each psychological state
Long vs. Short Performance	Comparative directional trade analysis across sentiment categories
Monthly PnL Trends	Time-series view of profitability alongside sentiment shifts
Top Traders	Accounts ranked by highest cumulative realized profit
Most Profitable Coins	Assets ranked by total Closed PnL contribution

Analytical Approach

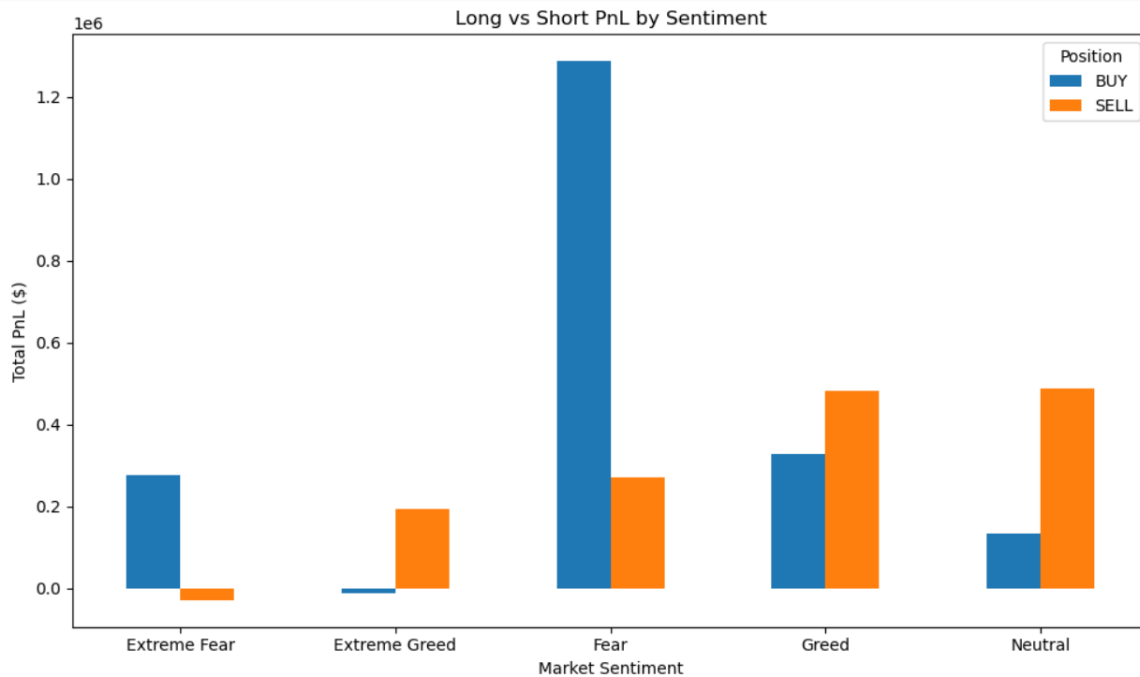
The project employs descriptive analytics with group-by aggregations, time-series tracking, and distribution analysis. Visualizations were produced using Matplotlib and Seaborn to surface patterns that raw tabular data would otherwise obscure.

Visualizations

Five core visualizations were developed to translate complex trading data into interpretable insights:

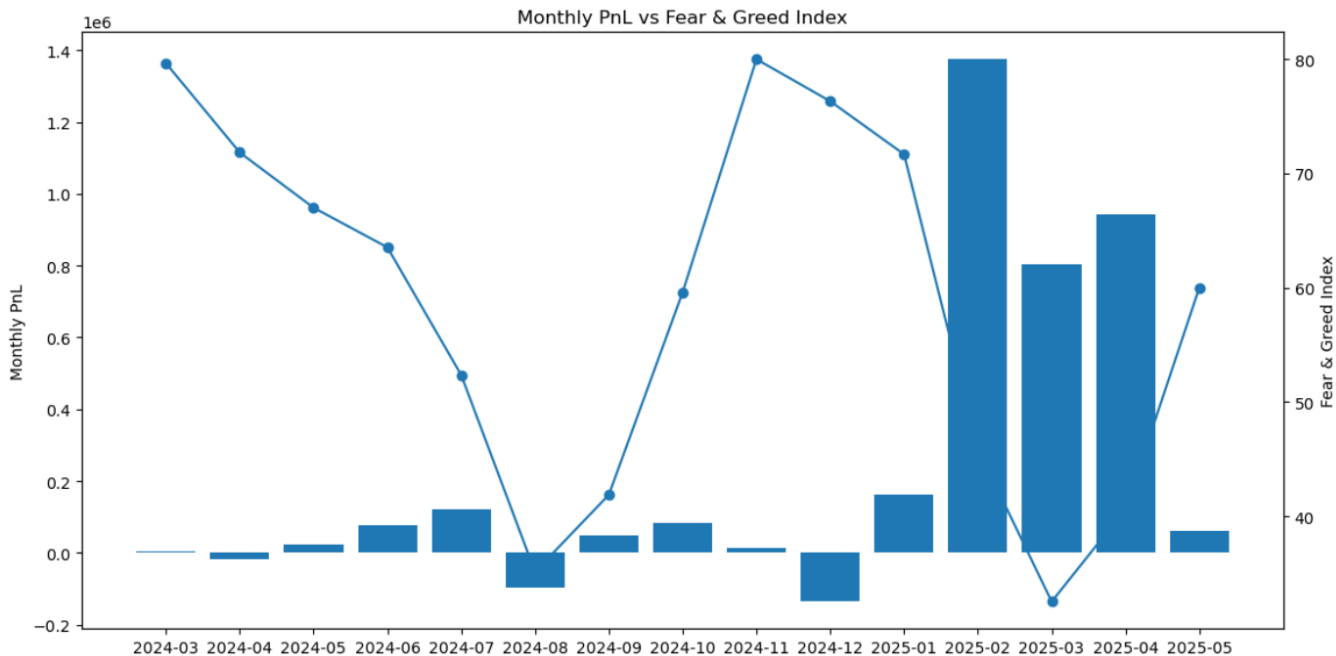
1. Long vs. Short PnL by Sentiment

A grouped bar chart displaying the profitability of long positions versus short positions across each sentiment category (Extreme Fear through Extreme Greed). This reveals whether directional bias should shift based on market psychology and which combination of direction and sentiment yields the best returns.



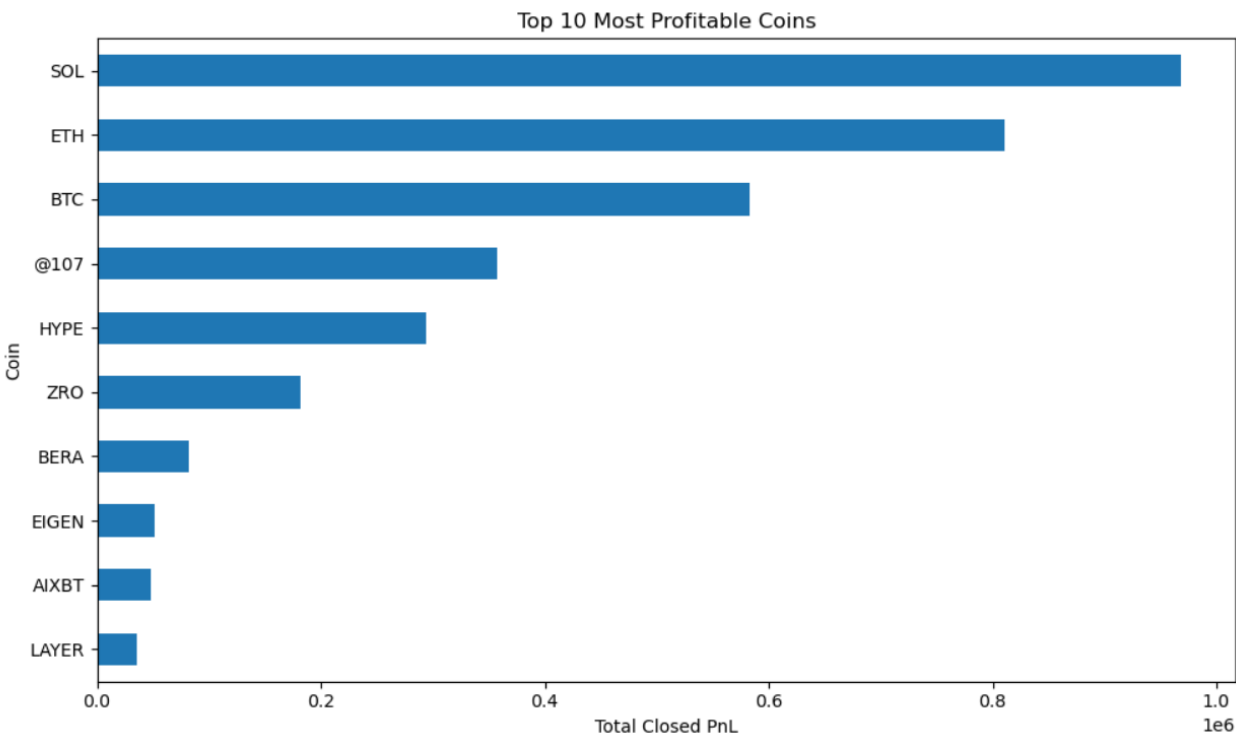
2. Monthly PnL vs. Fear & Greed Index

A dual-axis time-series plot tracking monthly aggregate trader profitability alongside the Fear & Greed sentiment trajectory. This visualization highlights temporal correlations between sentiment extremes and profit peaks or drawdown periods.



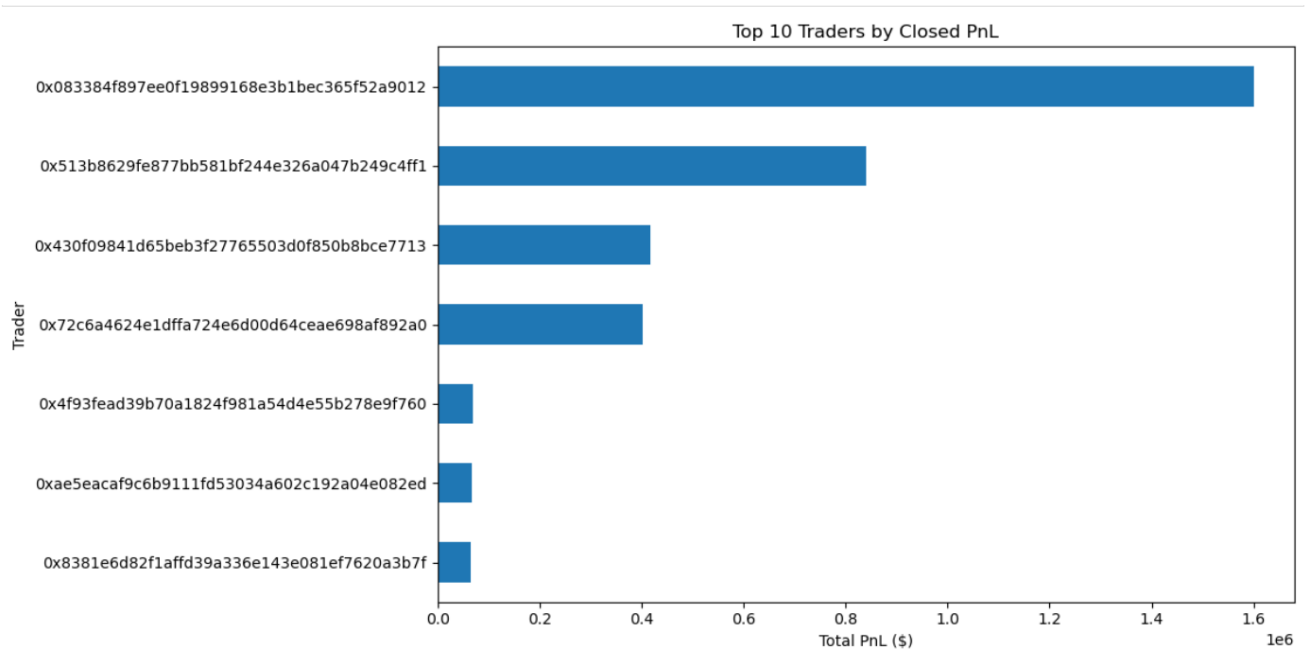
3. Top 10 Most Profitable Coins

A ranked horizontal bar chart identifying which cryptocurrency assets contributed the highest cumulative realized profits. This isolates high-performing instruments within the Hyperliquid ecosystem and informs asset selection strategy.



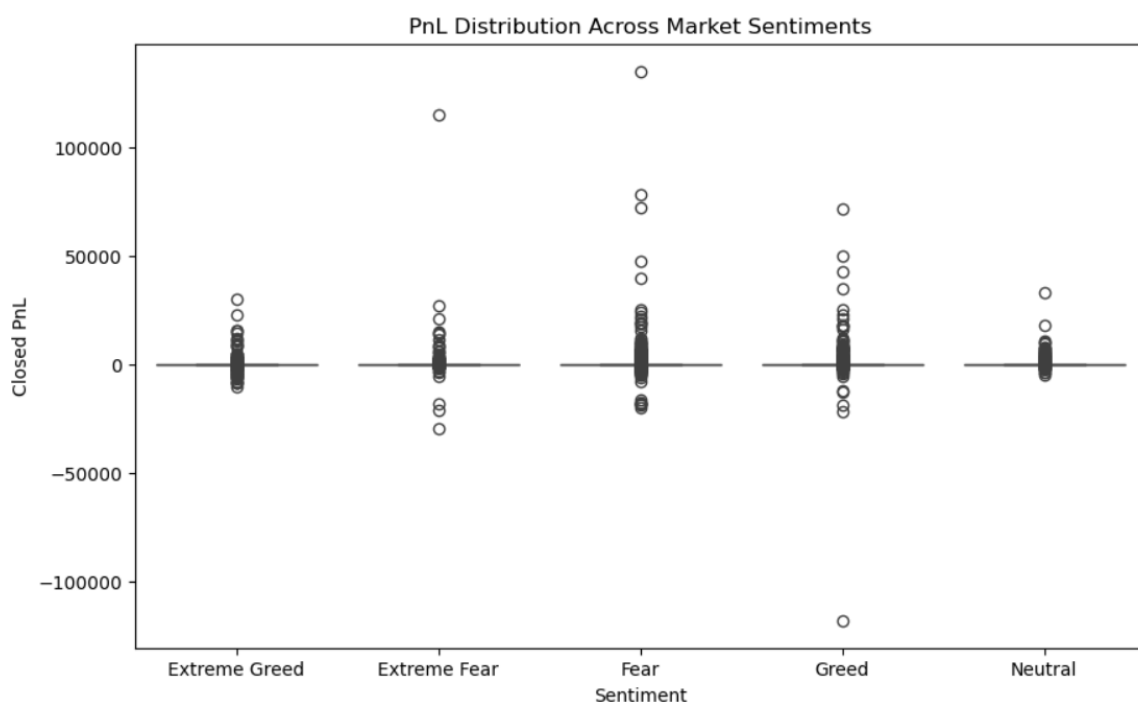
4. Top 10 Traders by Closed PnL

A leaderboard visualization ranking the top ten trader accounts by total realized profit. This chart exposes performance concentration — how much total profit is captured by elite accounts — and identifies standout trading behavior.



5. PnL Distribution by Market Sentiment

A box plot (or violin plot) showing the spread and volatility of profit/loss outcomes across each sentiment state. This illustrates not just average performance but also the risk and variability that traders face under different market conditions — making it crucial for risk management decisions.



Insights

1. Sentiment Influences Trading Performance

Trader profitability is not uniform across market conditions. Aggregate PnL varies significantly depending on whether the market is experiencing Fear, Greed, or Neutral sentiment. This confirms that psychological climate — not just price action alone — has a measurable impact on trading outcomes across the Hyperliquid platform.

2. Trade Direction Matters Under Different Conditions

Long and short positions exhibit divergent performance profiles under different sentiment regimes. Long positions tend to outperform during greed-driven rallies when bullish momentum is strong, while short positions may capture more value during fear-driven selloffs. The optimal directional bias is not static — it shifts meaningfully with market psychology.

3. Asset Concentration Drives Profit

A small subset of traded assets generates a disproportionately large share of total realized profits. This Pareto-style distribution suggests that concentrating capital on historically high-performing coins — rather than spreading exposure broadly — may significantly improve overall profitability on the platform.

4. Performance Concentration Among Top Traders

Top-performing traders account for a significant portion of aggregate profitability. This indicates a clear skill or strategy gap between elite and average participants. Studying the behavior of top accounts — including their preferred assets, sentiment-timing patterns, and position sizing — could yield replicable strategic insights.

5. Extreme Sentiment Creates Both Risk and Opportunity

Extreme Fear and Extreme Greed states are associated with distinct behavioral changes, including altered position sizing, increased trade frequency, and higher volatility in PnL distributions. While these psychological extremes carry elevated downside risk, they also represent periods of maximum opportunity for disciplined, strategy-driven traders.

Recommendations

1. Incorporate the Fear & Greed Index as a Systematic Signal

Use the Fear & Greed Index as a secondary confirmation layer alongside technical and fundamental analysis. Sentiment extremes often precede price reversals or momentum acceleration — making it a useful filter before entering or sizing positions.

2. Align Trade Direction with Market Psychology

Favor long-biased strategies during greed and neutral-upward sentiment phases. Shift to short-biased or defensive positioning during fear-dominated periods. The data suggests direction-sentiment alignment meaningfully impacts win rates and PnL.

3. Exercise Caution at Sentiment Extremes Before Scaling Exposure

Avoid large capital deployment when the index approaches Extreme Greed or Extreme Fear thresholds unless the trading strategy specifically targets those regimes. Volatility and PnL dispersion both increase dramatically at extremes.

4. Concentrate on High-Performing Assets

Focus trading activity on the top-performing coins identified in the analysis. Avoid spreading attention across low-conviction or historically unprofitable assets. A focused watchlist aligned with proven performers reduces noise and improves risk-adjusted returns.

5. Strengthen Risk Management During Extreme Sentiment Periods

Reduce position sizes and tighten stop-loss levels when the Fear & Greed Index hits extreme values. The PnL distribution analysis shows that outcome variability peaks during these periods — making disciplined risk management essential to capital preservation.

Conclusion

The PrimeTrade Market Sentiment Analysis demonstrates that Bitcoin market sentiment, captured through the Fear & Greed Index, has a measurable and actionable relationship with Hyperliquid trader performance. By cleaning, integrating, and visualizing two distinct datasets, the project reveals that profitability, trade direction, and risk exposure are all materially influenced by collective market psychology.

These findings provide a data-driven foundation for building sentiment-aware trading strategies. More broadly, the project underscores the value of behavioral analytics in cryptocurrency markets — where emotional cycles between fear and greed are frequent, quantifiable, and exploitable for traders with the right analytical framework.

Report generated from: [PrimeTrade-Sentiment-Trading-Analysis GitHub Repository](#) .